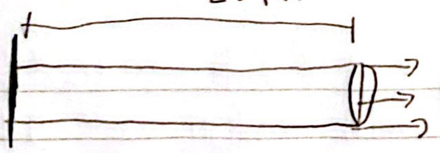
[redacted]

Elongation $= \frac{L_f - L_o}{L_o} \cdot 100$ $\qquad E = 8.5 \cdot 10^6$ psi

1. $E = \frac{\sigma}{e}$



$\boxed{\begin{array}{l} L = 1\text{ ft} \\ E = 8.5 \cdot 10^6 \text{ psi} \\ F = 500 \text{ lb} \end{array}}$ a.

$\delta = \frac{FL}{AE} \rightarrow A = \frac{FL}{\delta E} \rightarrow \frac{(500\text{ lb})(12\text{ in})}{(8.5 \cdot 10^6 \text{psi})(.009\text{ in})} \rightarrow$

$\rightarrow A = 0.07843 \text{ in}^2 \qquad A = \frac{\pi d^2}{4} \rightarrow \sqrt{\frac{A4}{\pi}} = \sqrt{\frac{.07843(4)}{\pi}} = d = .31601 \text{ in}$

$\delta = \frac{(500\text{ lb})(12\text{ in})}{(.31372 \text{ in}^2)(8.5 \cdot 10^6 \text{ psi})} = \boxed{.00225 \text{ in}}$

$d > .31601$

safety factor 2.00

$.31601 \cdot 2 = .63201$ in

$\frac{\pi(.63201)^2}{4} = .31372 \text{ in}^2$

$E = \frac{L_f - L_o}{L_o} = \frac{12.00225 - 12}{12} \cdot 100 = .019\%$ b.